

5 A transformation of the x - y plane is represented by the matrix $\begin{pmatrix} \cos \theta & 2 \sin \theta \\ 2 \sin \theta & -\cos \theta \end{pmatrix}$, where θ is a positive acute angle.

(i) Write down the image of the point $(2, 3)$ under this transformation. **[2]**

(ii) You are given that this image is the point $(a, 0)$. Find the value of a . **[5]**